Step 1: Understand the Data

```
from google.colab import files
uploaded = files.upload()
```

Choose Files data.csv
data.csv(text/csv) - 158 bytes, last modified: 2/24/2026 - 100% done
Saving data.csv to data.csv

```
import pandas as pd

df = pd.read_csv("data.csv")
df.head()
```

	name	age	city	category	sales	a	b
0	John	25	Delhi	A	200	5	10
1	Riya	22	Mumbai	B	120	2	3
2	Aman	30	Pune	A	250	7	8
3	Neha	27	Kolkata	C	300	4	6
4	Arjun	35	Jaipur	B	200	1	9

Next steps: Generate code with df New interactive sheet

```
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   name      5 non-null      object
 1   age       5 non-null      int64
 2   city      5 non-null      object
 3   category  5 non-null      object
 4   sales     5 non-null      int64
 5   a         5 non-null      int64
 6   b         5 non-null      int64
dtypes: int64(4), object(3)
memory usage: 412.0+ bytes
```

	age	sales	a	b
count	5.000000	5.000000	5.000000	5.000000
mean	27.800000	214.000000	3.800000	7.200000
std	4.969909	66.932802	2.387467	2.774887
min	22.000000	120.000000	1.000000	3.000000
25%	25.000000	200.000000	2.000000	6.000000
50%	27.000000	200.000000	4.000000	8.000000
75%	30.000000	250.000000	5.000000	9.000000
max	35.000000	300.000000	7.000000	10.000000

Data Cleaning

```
df.isnull().sum()
```

	0
name	0
age	0
city	0
category	0
sales	0
a	0
b	0

dtype: int64

for filling the missing values in numeric column

```
df['sales'] = df['sales'].fillna(df['sales'].mean())
```

Data Transformation

```
df_encoded = pd.get_dummies(df, columns=['city','category'])

df_encoded.head()
```

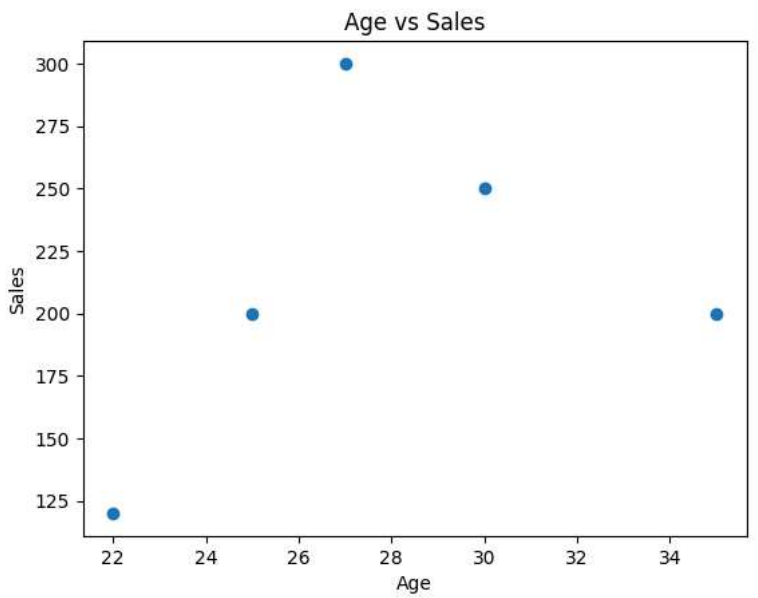
	name	age	sales	a	b	city_Delhi	city_Jaipur	city_Kolkata	city_Mumbai	city_Pune	category_A	category_B	category_C
0	John	25	200	5	10	True	False	False	False	False	True	False	False
1	Riya	22	120	2	3	False	False	False	True	False	False	True	False
2	Aman	30	250	7	8	False	False	False	False	True	True	False	False
3	Neha	27	300	4	6	False	False	True	False	False	False	False	True
4	Arjun	35	200	1	9	False	True	False	False	False	False	True	False

Next steps: Generate code with df_encoded New interactive sheet

Data Visualization

```
import matplotlib.pyplot as plt

df['sales'].hist()
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()
```

Data Visualization Part 2 :-

```
plt.scatter(df['age'], df['sales'])
plt.xlabel("Age")
plt.ylabel("Sales")
plt.title("Age vs Sales")
plt.show()
```